Bohan Zhang

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Education

University of Michigan
Ph.D. in Information Science — Aug 2022 - Present
M.S. in Computer Science and Engineering — Aug 2020 - May 2022
Research Interests: Causal Inference, Personalization, Machine Learning

The Ohio State University — Aug 2017 - May 2020
B.S. in Computer Science and Engineering, Minor in Linguistics

Xiamen University — Aug 2015 - July 2017
Major in Computer Science and Technology (Transferred out)

Publications and Working Drafts

1. **Bohan Zhang**, Anqi Ni, Yixin Wang, Paramveer Dhillon "Weightless Fine-Tuning: Personalizing LLMs via Logit-Space Transport." *COLM '26.*
2. **Bohan Zhang**, Jiaxuan Li, Ali Hortaçsu, Xiaoyang Ye, Victor Chernozhukov, Angelo Ni, Eddie Huang. "Agentic Economic Modeling." *ACL '26 Industry Track.* Link
3. Yachuan Liu, Xiaochun Wei*, Lin Shi*, Xinnuo Li*, **Bohan Zhang**, Paramveer Dhillon, Qiaozhu Mei "ExAnte: A Benchmark for Ex-Ante Inference in Large Language Models" *Main EACL '26.* [Link].
4. **Bohan Zhang***, Yachuan Liu*, Qiaozhu Mei, Paramveer Dhillon. "Exploring Group-Level Signals for Robust Many-Domain Generalization." *PAKDD '25.*
5. **Bohan Zhang**, Yixin Wang, Paramveer Dhillon. "Causal Inference for Human-Language Model Collaboration." *Main NAACL '24.* [Link].
6. Kenan Alkiek, **Bohan Zhang**, David Jurgens. "Classification without (Proper) Representation: Political Heterogeneity in Social Media and Its Implications for Classification and Behavioral Analysis." *Findings of ACL '22.* [Link].
7. **Bohan Zhang**, Prafulla Kumar Choubey, Ruihong Huang. "Predicting Sentence Deletions for Text Simplification Using a Functional Discourse Structure." *Main ACL '22.* [Link].
8. Yingzi Bu*, Ruoxi Gao*, **Bohan Zhang***, Luchen Zhang, Duxin Sun. "CoGT: Ensemble Machine Learning Method and its Application on JAK Inhibitor Discovery." *ACS Omega.* [Link].
9. **Bohan Zhang**, Chengke Bu, Paramveer Dhillon. "Who Owns the Text? Pairing Personas with Personalization Mitigates Ownership Loss in AI-Assisted Writing."
10. **Bohan Zhang**, Yixin Wang, Paramveer Dhillon. "Policy Learning with Text-Based Actions: A Causal Approach." [Link]
11. Yachuan Liu*, **Bohan Zhang***, Qiaozhu Mei, Paramveer Dhillon. "Improving Group Distributional Robustness by Learning to Rank." [Link].
12. Xiaoyang Ye, **Bohan Zhang**, Yudi Wang, Evan Peet. "Human-Algorithm Collaboration for Personalization: Evidence from Field and GenAI Experiments." *Amazon Economist Summit' 25*

Working Experiences

Applied Scientist Intern, Amazon — May 2026 - Now
- Developing a reasoning-distillation and GRPO fine-tuning pipeline that trains scalable reasoning LLMs to estimate how customer demand redistributes under catalog changes.
- Exploring RCT fine-tuning that aligns model outputs with treatment effects, grounding the LLM's counterfactual predictions in real causal signal.

Applied Scientist Intern, Amazon — Mar 2025 - Oct 2025
- Designed LLM-based framework aligning LLM-generated synthetic responses with small-scale human data for reliable economic inference.

- Validated the method in large-scale conjoint study and regional pricing RCTs, matching full-scale results using only 10% human samples.

Machine Learning Software Engineering, Initium.AI Inc. May 2021 - Aug 2021

- Develop and deploy action item detection model with limited labels.

Instructor Assistant, The Ohio State University Jan 2019 - May 2020

- Course: CSE 2421 - Introduction to Low-Level Programming and Computer Organization

Related Research Experiences

Research Assistant, Dhillon's Research Group

- Project 1: Dynamic sequence optimization in human-AI collaborative writing.
  - Collaborator: Yixin Wang
  - Goals: learn a dynamic causal policy – a sequence of treatments, which are texts generated by humans, to optimize the quality of texts in human-AI collaborative writing.
- Project 2: Personalized AI-Assisted Writing
  - Goals: Develop collaborative human-AI writing tools to enhance writing ownership.
- Project 3: Group distributional robustness; Data leakage in ex-ante inferences.
  - Collaborator: Qiaozhu Mei
  - Goals: 1) Improving the robustness by up-weighting groups according to their performance rank 2) Design generalizable meta-learners to learn the rank of groups directly. 3) Teaching LLMs to make ex-ante inferences (unlearning).

Research Assistant, Blablablab Lab Sep 2020 - Oct 2021

- Advisor: David Jurgens
- Project: Political Heterogeneity in Social Media and Its Implications for Behavioral Analysis.
- Goals: Understanding (1) how the choice in definition of a political user significantly influences behavioral analysis (2) how political users engage and interact with others (2) potential suspicious behavior of users, and (4) how users change their political leanings over time.

Research Assistant, NLP Lab at Texas A&M University May 2019 - Aug 2019

- Advisor: Ruihong Huang
- Project: Predicting sentence deletions in text simplification via a functional discourse structure
- Goals: 1) Exploring how a functional discourse structure correlated with sentence deletions in document-level simplification 2) Using the structure to improve predicting sentence deletions.

Honors and Service

ACL Student Volunteer Award
Dean's List of Distinguished Students All Semesters
Regular Reviewer for *CL, NeurIPS

Skills

Programming languages
Familiar with Python, C, Matlab; Have knowledge in C#, Java, JavaScript
Programming Tools
Pytorch, Linux, OpenGL
English Proficiency
TOEFL 107 (R:27, L:27, S:26, W:27)
GRE: V 157, Q 169